

Quiz 9

I. $\mathcal{H}_a: \frac{x^2}{a^2} - \frac{y^2}{4} = 1, a > 0$. (hyperbola)

✓ 1) $a = ?$ s.t. asymptotes have an angle of 45° with the x-axis.

canonical hyperbola: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, b = 2$

asymptotes: $\boxed{y = \pm \frac{b}{a}x} = \pm \frac{2}{a}x \Rightarrow m = \pm \frac{2}{a}$ (slope)

tangent lines of slope K : $y = Kx \pm \sqrt{a^2 K^2 - b^2}$

line making 45° angle with the x-axis has slope $\tan 45^\circ = 1 = m$

$\Rightarrow \pm \frac{2}{a} = 1 \Rightarrow a = 2 > 0$.

✓ 2) $a = 1$, eq. of the tangent lines to $\mathcal{H}_a$ that pass through $Q(0, 1)$.

$$\begin{cases} \mathcal{H}_1: x^2 - \frac{y^2}{4} = 1 \\ \text{line through } Q(0, 1): y = Kx + 1 \end{cases}$$

$\Rightarrow x^2 - \frac{(Kx+1)^2}{4} = 1 \quad | \cdot 4$

$\Rightarrow 4x^2 - (Kx+1)^2 = 4 \Rightarrow 4x^2 - K^2x^2 - 2Kx - 1 = 4 \Rightarrow$

$\Rightarrow (4-K^2)x^2 - 2Kx - 5 = 0$.

$\Delta = (-2K)^2 - 4(4-K^2)(-5) = 4K^2 + 20(4-K^2) = 0$ ↙ for tangency

$\Rightarrow -16K^2 + 80 = 0 \Rightarrow 16K^2 = 80 \Rightarrow K^2 = 5$

$\Rightarrow K = \pm \sqrt{5}$

$\Rightarrow y = \sqrt{5}x + 1$

$\Rightarrow y = -\sqrt{5}x + 1$

✓ 3) $a = ?$ s.t. $\mathcal{H}_a$ is tangent to $l: x - y = 1$.

$\Rightarrow y = x - 1 \Rightarrow \underline{K = 1}$ (slope)

$y = Kx + m = x - 1$

$\Rightarrow \begin{cases} K = 1 \\ m = -1 \end{cases}$

$\boxed{y = Kx \pm \sqrt{a^2 K^2 - b^2}}$ | tangent line form

$$\begin{cases} y = x \pm \sqrt{a^2 - b^2} \\ y = x - 1 \end{cases} \Rightarrow -\sqrt{a^2 - b^2} = -1 \Rightarrow \sqrt{a^2 - b^2} = 1 \xrightarrow{b=2} \Rightarrow \sqrt{a^2 - 4} = 1 \quad |^2 \Rightarrow a^2 - 4 = 1 \Rightarrow a = \pm \sqrt{5} \xrightarrow{a>0} \Rightarrow a = \sqrt{5}$$

II: $\mathcal{H}: x^2 - y^2 = 1$ (hyperbola)

✓ 4) $d(F_1, F_2)$

canonical hyperbola: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $a=1, b=1$

$c^2 = a^2 + b^2 = 1 + 1 = 2 \Rightarrow c = \pm\sqrt{2}$

$F_1(\sqrt{2}, 0), F_2(-\sqrt{2}, 0)$

$d(F_1, F_2) = \sqrt{(-\sqrt{2} - \sqrt{2})^2 - (0 - 0)^2} = \sqrt{(-2\sqrt{2})^2} = \sqrt{8} = 2\sqrt{2}$

✓ 5) tangent lines to $\mathcal{H}$ that are parallel to the line $l: 5x - 4y - 1 = 0$

$\Rightarrow$ slope: $4y = 5x - 1$

$y = \frac{5}{4}x - \frac{1}{4} \Rightarrow \text{slope} = \frac{5}{4} = k$

$y = kx \pm \sqrt{k^2 - 1}$

$y = \frac{5}{4}x \pm \sqrt{\frac{25}{16} - 1} = \frac{5}{4}x \pm \sqrt{\frac{9}{16}} = \frac{5}{4}x \pm \frac{3}{4}$

$y = \frac{5}{4}x + \frac{3}{4}$

$y = \frac{5}{4}x - \frac{3}{4}$

6) Equation for the hyperbola obtained by rotating $\mathcal{H}$ counterclockwise by 45° . ($\theta = 45^\circ$)

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$

$= \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} \frac{X-Y}{\sqrt{2}} \\ \frac{X+Y}{\sqrt{2}} \end{pmatrix}$

$\mathcal{H}: x^2 - y^2 = 1$

$\Rightarrow \begin{cases} x = \frac{X-Y}{\sqrt{2}} \Rightarrow \sqrt{2}x = X-Y \Rightarrow X = \sqrt{2}x + Y \\ y = \frac{X+Y}{\sqrt{2}} \Rightarrow \sqrt{2}y = X+Y \end{cases}$

$(+)$
 $x+y = \frac{X-Y}{\sqrt{2}} + \frac{X+Y}{\sqrt{2}} = \frac{2X}{\sqrt{2}} = \sqrt{2}X$

$\Rightarrow X = \frac{x+y}{\sqrt{2}}$

$(-)$ $\Rightarrow y-x = \frac{X+Y}{\sqrt{2}} - \frac{X-Y}{\sqrt{2}} = \sqrt{2}Y$

$\Rightarrow Y = \frac{y-x}{\sqrt{2}}$

$$x^2 - y^2 = 1$$

$$\Rightarrow \left(\frac{x+y}{\sqrt{2}} \right)^2 - \left(\frac{-x+y}{\sqrt{2}} \right)^2 = 1$$

$$\Rightarrow \frac{(x+y)^2 - (-x+y)^2}{\sqrt{2}^2} = 1$$

$$\Rightarrow x^2 + 2xy + y^2 - (x^2 - 2xy + y^2) = \sqrt{2}^2$$

$$\Rightarrow 4xy = \sqrt{2}^2 \Rightarrow xy = \frac{1}{2} \quad (\text{rotated hyperbola})$$

4. $\lambda = ?$ s.t. $P(5, \lambda)$ does not lie on any tangent line of $\mathcal{H}$ (hyperbola).

$$\mathcal{H}: x^2 - y^2 = 1$$

tangent line of slope k : $y = kx + m$

$$x^2 - (kx + m)^2 = 1$$

$$y^2 = k^2 x^2 + m^2$$

$$\Rightarrow x^2 - (k^2 x^2 + 2mkx + m^2) = 1$$

$$\Rightarrow (1 - k^2)x^2 - 2mkx - m^2 = 1$$

$$\Delta = 0 \dots$$

$$\boxed{y = kx \pm \sqrt{a^2 k^2 - b^2}} \quad a = b = 1$$

$$y = kx \pm \underbrace{\sqrt{k^2 - 1}}_m \Rightarrow y = kx + m = kx \pm \sqrt{k^2 - 1}$$

$$\Leftrightarrow m = \pm \sqrt{k^2 - 1} \quad |^2$$

$$\Rightarrow \underline{m^2 = k^2 - 1}$$

$$y = kx + m$$

$$P(5, \lambda) \uparrow$$

$$\Rightarrow \lambda = 5k + m \Rightarrow m = \lambda - 5k$$

$$(\lambda - 5k)^2 = k^2 - 1 \Rightarrow \lambda^2 - 10\lambda k + 25k^2 = k^2 - 1 \Rightarrow$$

$$\Rightarrow \underline{24k^2 - 10\lambda k + \lambda^2 + 1 = 0}$$

$\exists$ at least one tangent $\Leftrightarrow \Delta \geq 0$

$$(-10\lambda)^2 - 4 \cdot 24 \cdot (\lambda^2 + 1) \geq 0$$

$$100\lambda^2 - 96\lambda^2 - 96 \geq 0$$

$$4\lambda^2 - 96 \geq 0$$

$$\lambda^2 \geq 24$$

$\exists$ tangents $\Leftrightarrow |\lambda| \geq 2\sqrt{6}$

$P(5, \lambda)$ does not lie on tangent $\Leftrightarrow |\lambda| < 2\sqrt{6}$

$$\Rightarrow \lambda \in (-2\sqrt{6}, 2\sqrt{6})$$

III. $P: y^2 = 2x$ (parabola)

✓ 8) eq. for directrix of P .

$$\boxed{P: y^2 = 2px \rightarrow \text{directrix} : x = -\frac{p}{2}}$$

$$p=1 \Rightarrow x = -\frac{1}{2}$$

✓ 9) tangent lines to P that have a 45° angle with the x -axis.

$$K = \tan 45^\circ = 1.$$

$$\boxed{y = Kx + \frac{p}{2K}} \text{ tangent of slope } K \text{ for } P$$

$$p=1, K=1$$

$$\hookrightarrow y = x + \frac{1}{2 \cdot 1} = x + \frac{1}{2}$$

10) $\alpha \rightarrow$ angle between x -axis and tangent line to P at $P(2, -2) \in P$
 $\cos \alpha = ?$

$$\boxed{yy_0 = p(x+x_0)} \quad P(2, -2); x_0=2, y_0=-2, p=1.$$

$$\Rightarrow -2y = x+2 \Rightarrow x+2y+2=0 \Rightarrow 2y = -2-x \Rightarrow y = \underbrace{\left(-\frac{1}{2}\right)}_K x - 1$$

$$\Rightarrow K = -\frac{1}{2}$$

$$\tan \alpha = -\frac{1}{2} \Rightarrow \alpha = \dots$$

$$\cos \alpha = \frac{\langle v, i \rangle}{|v| \cdot |i|} = \frac{2}{\sqrt{5} \cdot \sqrt{1}} = \frac{2}{\sqrt{5}} \quad \begin{array}{c} \text{suplem.} \\ \text{angles} \end{array} \Rightarrow \pm \frac{2}{\sqrt{5}}$$

dir. vector of the tangent: $v = (2, -1)$

dir. vector of the x -axis: $i = (1, 0)$

$$\langle v, i \rangle = v^T \cdot i = (2, -1) \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 2.$$

$$|v| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$|i| = \sqrt{1^2 + 0^2} = \sqrt{1}.$$

$$\cancel{\begin{pmatrix} 2 \\ -1 \end{pmatrix}} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} =$$

11) $a = ?$ st. $l_a: x - ay + 2 = 0$ is tangent to P .

$$\Rightarrow ay = x + 2 \Rightarrow y = \left(\frac{1}{a}\right)x + \frac{2}{a} \Rightarrow K = \frac{1}{a}$$

$$\boxed{y = Kx + \frac{p}{2K}} \quad p=1, K=\frac{1}{a} \Rightarrow y = Kx + \frac{1}{2K} = \frac{1}{a}x + \frac{2}{a}$$

$$\Rightarrow y = \frac{1}{a}x + \frac{1}{2 \cdot \frac{1}{a}}$$

$$\begin{aligned} \Leftrightarrow \begin{cases} K = \frac{1}{a} \\ \frac{1}{2K} = \frac{2}{a} \end{cases} &\Rightarrow a = 4 \cdot \frac{1}{a} \\ &\Rightarrow a^2 = 4 \Rightarrow a = \pm 2. \end{aligned}$$

Quiz 9

I. $\mathcal{H}a: \frac{x^2}{a^2} - \frac{y^2}{4} = 1, a > 0.$

1) $a = ?$ s.t. asymptotes, $\theta = 45^\circ \Rightarrow$ slope $k = \tan 45^\circ = 1.$

$$a = a > 0$$

$$b = 2$$

asymptotes of a hyperbola: $y = \pm \frac{b}{a} x = \pm \frac{2}{a} x$

$$\Rightarrow k = \pm \frac{2}{a} \text{ (slope)}$$

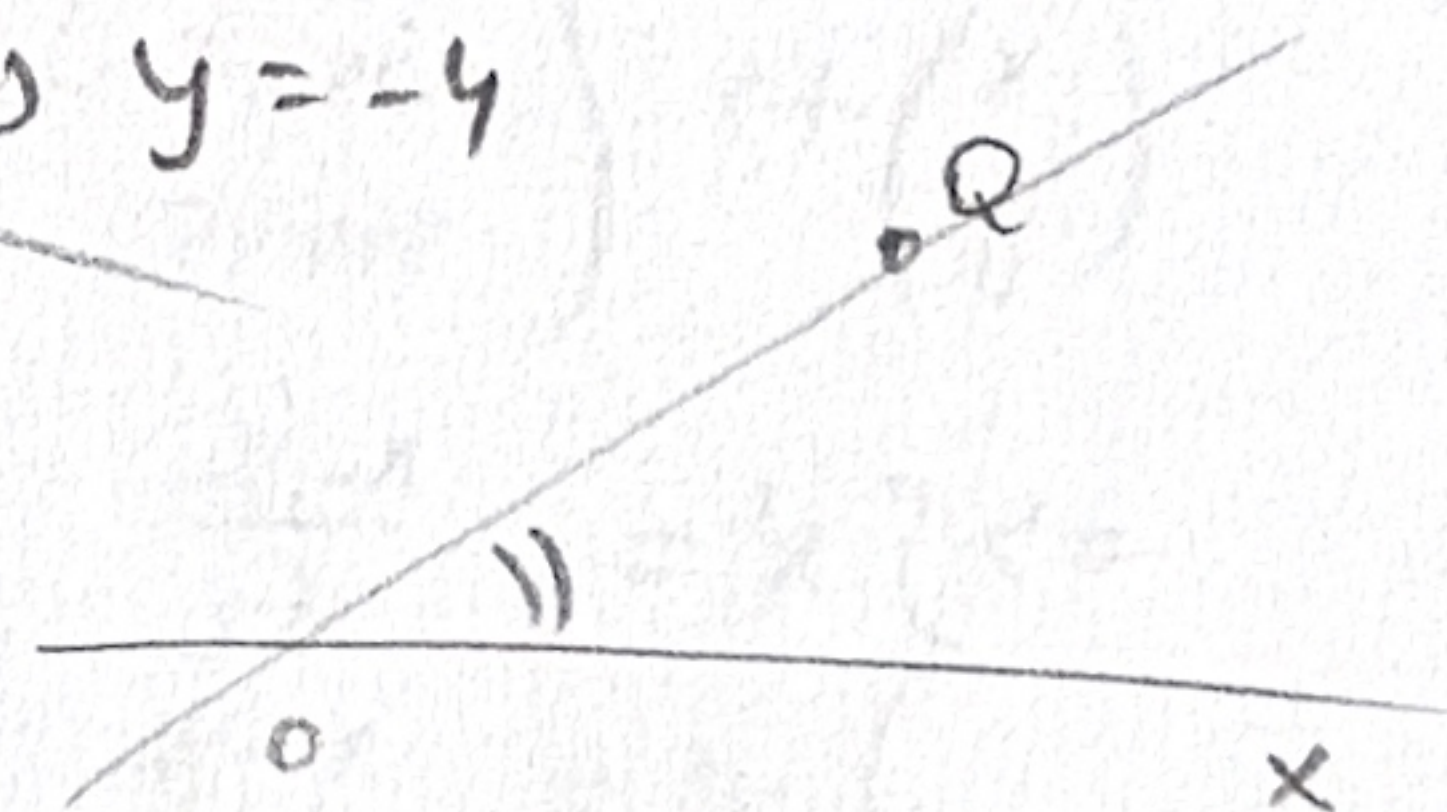
$$\Rightarrow k = \pm \frac{2}{a} = 1 \Rightarrow a = \pm 2 \quad \left. \begin{array}{l} a > 0 \end{array} \right\} \Rightarrow \underline{a = 2}.$$

! 2) $a = 1$, tg lines passing through $Q(0,1)$

~~$\mathcal{H}p: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$~~

~~$\Rightarrow \frac{0}{1} - \frac{1}{4} = 1 \Rightarrow -y = 4 \Rightarrow y = -4$~~

~~$y = kx \pm \sqrt{a^2 k^2 - b^2}$ not useful here~~



$\mathcal{H}: \frac{x^2}{1^2} - \frac{y^2}{2^2} = 1$

$y = kx + 1 \rightarrow$ line passing through $Q(0,1) \in \mathcal{H} \rightarrow 1 = k \cdot 0 + 1 \Rightarrow k = 1.$

$x^2 - \frac{(kx+1)^2}{4} = 1 \quad | \cdot 4 \Rightarrow \dots \Delta = 4k^2 + 20(4-k^2) \text{ impose } = 0$

$$\Rightarrow k^2 = 5 \Rightarrow k = \pm \sqrt{5}$$

$y_1 = \sqrt{5}x + \sqrt{1+5} = \sqrt{5}x + 1$

$y_2 = -\sqrt{5}x + 1$

$\pm \sqrt{a^2 k^2 - b^2} = m$

$\Rightarrow \sqrt{5}x - y + 1 = 0 \text{ and } -\sqrt{5}x - y + 1 = 0.$

3) $a = ?$ s.t. $\mathcal{H}a$ tg to $l: x - y = 1 \Rightarrow y = x - 1 \Rightarrow k = 1$ (slope).

$y = kx \pm \sqrt{a^2 k^2 - b^2} \quad \left(\frac{k=1}{\quad} \right) x \pm \sqrt{a^2 - b^2} = y$

$\Rightarrow x \pm \sqrt{a^2 - b^2} = x - 1 \quad | -x$

$\Rightarrow \pm \sqrt{a^2 - b^2} = -1 \quad |^2 \Rightarrow a^2 - 4 = 1 \Rightarrow a^2 = 5$

$b = 2$

$\Rightarrow a = \pm \sqrt{5} \xrightarrow{a > 0} a = \sqrt{5}.$

II. $\mathcal{H}: x^2 - y^2 = 1 \quad a=1, b=1$

4) $d(F_1, F_2) = ?$

$c^2 = a^2 + b^2 = 1 + 1 = 2 \Rightarrow c = \pm\sqrt{2} \rightarrow F_1(\sqrt{2}, 0)$
 $\rightarrow F_2(-\sqrt{2}, 0)$

$d(F_1, F_2) = \sqrt{(-\sqrt{2} - \sqrt{2})^2 + 0} = \sqrt{8} = 2\sqrt{2}$

5) $\text{tg lines to } \mathcal{H} \text{ that are } \parallel \ell: 5x - 4y - 1 = 0$

$y = kx \pm \sqrt{a^2 k^2 - b^2}$

$\Rightarrow 4y = 5x - 1$

$y = \frac{5}{4}x \pm \sqrt{\frac{25}{16} - 1}$

$y = \left(\frac{5}{4}\right)x - \frac{1}{4}$
 $\hookrightarrow k$

$y = \frac{5}{4}x \pm \sqrt{\frac{9}{16}} \quad \checkmark$

6) Hyp? $\mathcal{H}$ rot. by $+45^\circ = \theta$

$\begin{pmatrix} x' \\ y' \end{pmatrix} \mapsto \begin{pmatrix} \cos 45 & -\sin 45 \\ \sin 45 & \cos 45 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$\Rightarrow \begin{cases} x' = \frac{x-y}{\sqrt{2}} \\ y' = \frac{x+y}{\sqrt{2}} \end{cases}$

$\Rightarrow \dots \Rightarrow x = \frac{x' + y'}{\sqrt{2}}$

$\Rightarrow y = \frac{y' - x'}{\sqrt{2}}$

$x'^2 - y'^2 = 1$

$\Rightarrow \left(\frac{x' + y'}{\sqrt{2}}\right)^2 - \left(\frac{y' - x'}{\sqrt{2}}\right)^2 = 1$

$\Rightarrow xy = \frac{1}{2}$

7) $\lambda = ?$ st. $\mathcal{P}(5, \lambda) \notin \text{tg. line of } \mathcal{H}$

$\boxed{y = kx \pm \sqrt{a^2 k^2 - b^2}} = kx \pm \sqrt{k^2 - 1} = kx + m$

(2) $m = \pm \sqrt{k^2 - 1}$

(2) $m^2 = k^2 - 1 \Rightarrow (5 - 5k)^2 = k^2 - 1$

$y = kx + m$
 $\mathcal{P}(5, \lambda) \uparrow \Rightarrow \lambda = 5k + m$

$\Rightarrow m = \lambda - 5k$

$\Rightarrow \dots \Delta \geq 0$

$\Rightarrow \lambda^2 \geq 24$

$\exists \text{tg}(\Rightarrow |\lambda| \geq 2\sqrt{6}$
 $\nexists \text{tg}(\Rightarrow |\lambda| < 2\sqrt{6}$

III. $\mathcal{P}: y^2 = 2x, p=1$

8) $x = -\frac{p}{2}$ directrix. $\Rightarrow x = -\frac{1}{2}$
 $p=1$ (here)

9) tg lines to $\mathcal{P}$, 45° w Ox .
 slope $= k = \tan 45^\circ = 1$, $p=1$

$$\boxed{y = kx + \frac{p}{2k}} = kx + \frac{1}{2}$$

10) α $\widehat{tg}$, $x = \alpha$ $\cos \alpha = ?$
 $\hookrightarrow$ tg at $P(2, -2) \in \mathcal{P}$.

$$y - y_0 = p(x + x_0)$$

$$-2y = x + 2$$

$$\Rightarrow y = \underbrace{\left(-\frac{1}{2}\right)}_k x - 1$$

$$\tan \alpha = k = -\frac{1}{2} \Rightarrow \alpha = ?$$

$$\cos \alpha = \frac{\langle v, i \rangle}{|v| \cdot |i|} = \frac{v \cdot i}{\sqrt{5} \cdot \sqrt{1}} = \frac{2}{\sqrt{5}} \Rightarrow \pm \frac{2}{\sqrt{5}} \text{ (suplem)}$$

$$N = (2, -1) \text{ dir. vector of } x + 2y + 2 = 0$$

$$(1, 2) \cdot (2, -1) = 2 - 2 = 0$$

$$i = (1, 0) \text{ dir. vector of } Ox.$$

11) $a = ?$ s.t. $l_a: x - ay + 2 = 0$ is tg to $\mathcal{P}$.

$$\Rightarrow ay = x + 2 \Rightarrow y = \left(\frac{1}{a}\right)x + \frac{2}{a}$$

$$\boxed{y = kx + \frac{p}{2k}} = \frac{1}{a} \cdot x + \frac{1}{2k \frac{1}{a}} = \frac{1}{a} \cdot x + \frac{a}{2}$$

$$\Rightarrow a = \pm 2$$